HEADSTART: Forecasting health-system crises from published evidence when local outcome data are scarce

1. Summary

At the start of a health crisis, many useful inputs are already there. Weather, air quality, calendar effects, wastewater and public surveillance data come from public feeds or established partners. What is missing is not this context. It is local, timely and reliable data on the outcome of interest for the episode that is starting. Past local records may be long, but they do not say how the new episode will affect that outcome. The model has to estimate this relationship early and update it as outcomes arrive.

HEADSTART asks whether published evidence from other settings and past events can improve forecasts before enough outcomes from the current episode are available, and whether the method can detect when that evidence is misleading. Published studies help identify plausible parameter values, relevant explanatory variables and their lags, suitable model forms, and outcome definitions and warning thresholds. A prior is a probability distribution for model parameters before it is updated with outcomes from the current episode. It represents uncertainty rather than treating a published estimate as exact. HEADSTART will extract this information, measure extraction error and assess how relevant each finding is to the local setting. The main focus is on comparing an evidence-informed strategy to a reference strategy. Both employ the same available context feeds, previous local outcomes, and the outcomes observed so far in the current episode. Only the evidence-informed strategy uses information extracted from published studies to refine its parameter distributions and model choices.

The work has four linked parts. First, I will compare automated extraction with a manual benchmark and measure how extraction errors change the uncertainty of the model inputs. Second, I will build a model that forecasts the chosen outcome and expresses it as ordered levels of pressure on the health system, with the number of levels and their thresholds taken first from the evidence and fixed before any test. Third, I will replay past events using only the information available at the time and compare the two strategies. Fourth, I will test whether differences in forecast quality would change operational decisions.

The tests take place in Geneva and cover three types of crisis with different causes. The main test uses respiratory epidemics, for which published evidence is comparatively rich. The second uses heat events. The third, in year four, uses legionellosis and a linked case-environment dataset to forecast case incidence. Before any clinical data are touched, the methods are validated on open Swiss series and released as a public benchmark. If published evidence improves forecasts, the project shows when and by how much. If it brings nothing or causes harm, the result says where it should not be used. For the crisis types that pass validation and that the services authorise, forecasts run on a daily dashboard in observation mode and are scored as outcomes arrive. The dashboard does not direct clinical care.

I trained as an econometrician and spent eleven years in quantitative finance on rare-event risk before a doctorate in biomedical sciences at Geneva. I conceived and built LiteRev with Aziza Merzouki and Olivia Keiser and extended it into LiteRev-Evidence, the prototype this project builds on. Through GESICA and GeoAI4EI (Horizon Europe) I work with Geneva's surveillance data and with European epidemic-intelligence partners. HEADSTART builds on this work to test a distinct question, in a programme I will lead.

2. Proposed research

2.1 Current state of research in the field

2.1.1 Forecasts need outcome history as well as context data

Syndromic surveillance detects departures from expected baselines [1, 2], and ensembles of forecasting models perform strongly when outcome data are rich [3, 4]. Emergency dispatch records move ahead of confirmed surveillance. The autumn wave of the 2009 A(H1N1) pandemic was seen eight days early in three European regions [5], long series of emergency calls track influenza-like illness [6], and a systematic review maps the field [7].

The usual problem is not a lack of context data. Weather, air quality, calendar effects, wastewater and some surveillance indicators are available in near real time and can be linked automatically to a forecasting model. What is missing at the start is data on the outcome itself. A new episode often differs from earlier ones in the population affected, the drivers and the severity, so the link between context and outcome has to be checked again as outcomes arrive.

Google Flu Trends shows why indirect indicators cannot replace this check. It overestimated influenza by more than a factor of two [8]. Published findings may help, provided their uncertainty and their relevance to the local setting are assessed.

2.1.2 Published studies can inform forecasts by assessing uncertainty and relevance

Statistics has methods for using earlier studies in a new setting while keeping their uncertainty and judging their relevance. These include power priors [9], commensurate priors [10] and robust meta-analytic-predictive priors [11]. Informative priors have also been used to forecast diseases with little local data [12]. What has not been shown is whether an automated pipeline can extract published evidence, keep its uncertainty, judge its relevance to the new setting and improve forecasts of local outcomes at the start of a crisis.

Published studies offer more than parameter values. They suggest which variables and time lags matter, which types of models work, and how to define outcomes and warning thresholds. In some fields this evidence is kept up to date in living systematic reviews [13]. To my knowledge, these different forms of evidence have never been combined and tested in a real forecasting setting to make up for missing outcome data.

Automated extraction makes large-scale evidence synthesis possible, but numbers are extracted less reliably than categories (reported accuracy roughly 47–88% against 74–96%), and omissions make up a large share of the errors [14]. HEADSTART tests whether these errors understate uncertainty in a systematic way and whether an explicit measurement-error model can correct the distributions.

Even a correctly extracted estimate may not apply locally, because populations, case definitions, service organisation and policy differ. Formal methods to assess transferability exist [15–17], but they have not been combined with automated extraction, and their effect on forecasts at the start of a crisis has not been measured.

2.1.3 Escalation of the demand for the service should also be assessed

Emergency services need to know not only how many calls or visits to expect, but whether demand is heading for a level that calls for action. Surveillance research has used hidden Markov models to separate epidemic from non-epidemic periods [18, 19], and extreme-value methods describe rare peaks in demand [20], including hospital congestion in Switzerland [21]. HEADSTART expresses its forecasts as ordered levels of pressure on the system, so that all candidate models can be compared on the same scale. The number of levels and their thresholds are taken first from the published definitions of

outcomes and warning thresholds, and four levels (routine, elevated, strained, critical) are the default only where the literature is silent. Transmission models [22] are considered only for infectious crises, and heat is modelled through exposure and response. Rises in variance and in day-to-day correlation, which precede some transitions [23] and have been studied in epidemics [24–26], are tested as possible extra warning signs, not assumed to be reliable.

2.1.4 Forecast quality is important when it affects a decision

A forecast is useful when it changes a decision for a good reason. Proper scoring rules judge probability forecasts, including their calibration [27], and decision analysis judges them by their consequences [28]. Heat thresholds built on mortality need not match those for service demand [29], and the link between air pollution and asthma emergencies varies between places [30]. HEADSTART therefore scores forecasts first statistically, then against the costs of wrong decisions as stated by emergency-service experts.

2.1.5 The gap, evaluating evidence in a different context

Existing efforts include the ECDC RespiCast and CDC FluSight forecasting hubs, the GeoAI4EI project (Horizon Europe) and the GESICA project (France and Switzerland). HEADSTART asks a different question. When local outcome data are still few, does evidence from earlier studies improve automated forecasts of what a health service will face? It answers with a comparison registered in advance, the same data streams for both strategies, and a replay of the past that excludes later information.

2.2 Current state of personal research and competences required for the project

2.2.1 From econometrics and finance to public health

I studied quantitative economics and econometrics (MSc and DEA, University of Geneva), then worked for eleven years in quantitative finance on risk, extreme values, regime models and stress testing, and became a certified Financial Risk Manager (FRM). This is where my practice of rare-event forecasting, regime-switching models and Bayesian modelling under misspecification comes from. In 2019 I joined the Institute of Global Health and completed a doctorate in biomedical sciences (defended 18 December 2023).

2.2.2 Doctorate, prediction under sparse information and LiteRev

In [31] I predicted individual HIV status from social and behavioural characteristics across East and Southern Africa and showed where the models transferred between countries, in a line of work shared with [32] and [33]. This work gave me direct experience of when a model learnt in one setting holds in another.

I conceived and built LiteRev [34] with Aziza Merzouki (PhD, computer science) and Prof. Olivia Keiser. The tool runs systematic searches from plain-language or Boolean queries, removes duplicates and maps the literature. LiteRev was funded by UNIGE (CHF 30,000), the Fondation Ernst et Lucie Schmidheiny (CHF 10,000), Venture Kick (CHF 10,000) and Mimosa (CHF 20,000). The GESICA systematic review of AI in emergency medical services [7], of which I am a co-author, uses LiteRev in its methods.

2.2.3 Outbreak and health-system modelling in Switzerland

In [35] I compared the clinical severity of Delta and Omicron sub-lineages in a Swiss university hospital. [36] built age-structured scenarios of the Swiss SARS-CoV-2 epidemic to support decisions under time pressure. I also contributed to WHO African Region reporting and to seroprevalence estimates [37]. As last author of [38] (accepted), I had the senior role in the statistical analysis of the classification of self-reported cardiovascular disease across harmonised WHO STEPS surveys.

2.2.4 Since 2024, LiteRev-Evidence, GESICA and GeoAI4EI

Since 2024 I have extended LiteRev into LiteRev-Evidence, a prototype for extracting numbers from studies and turning them into models. Today it searches and deduplicates the literature, records screening decisions with PRISMA accounting, extracts study details and PICO elements (population, intervention, comparator and outcome), keeps the source and a certainty grade for each number, and pools the numbers into distributions of plausible parameter values. It also suggests explanatory variables with their supporting studies, lists candidate outcomes, fits models of several families with time-series validation, and builds compartmental model parts from published parameter ranges. Evidence updates are reviewed before they are applied. All of this exists as a prototype only. None of it has been verified, measured or validated yet, and that is the work of this grant.

GESICA is a large Franco-Swiss consortium. In it, the Geneva group led by Prof. Olivia Keiser works on infectious-disease crises, starting with influenza, COVID-19 and RSV. My contribution was to list the variables and data sources for these diseases, and for heat waves, in an inventory of 28 surveillance sources (23 on diseases and 5 on the environment), each described by coverage, resolution, latency, access and quality. HEADSTART uses this inventory. Within GeoAI4EI (Horizon Europe, 2026–2030) I contribute to the European epidemic-intelligence toolbox. The question of HEADSTART grew out of the meeting between LiteRev and these two projects on predicting health crises.

The prototype lowers the development risk, since search, extraction and model selection already run in a first form. The grant is about the open scientific questions, whether the extracted evidence is reliable enough, whether it transfers to the local setting, and whether it improves forecasts under the registered comparison.

2.2.5 The legionellosis study

I carry out the full analysis of a cantonal study already under way (BASEC 2026-00324, ethics granted, PI Prof. Olivia Keiser) that links confirmed legionellosis cases in Geneva to individual domestic hot-water installations. These linked data are the basis of the third test, on legionellosis case incidence, in year four.

Together these steps give me the methods for forecasting rare events, the evidence pipeline, and the Geneva and public-health context that HEADSTART needs.

2.3 Detailed research plan

2.3.1 Objectives and hypotheses

The central hypothesis is that published evidence can help estimate the relationship between the inputs and the outcome of interest, provided its uncertainty and local relevance are represented.

	Role	Statement
H1	Evidence quality	Does automated extraction preserve the uncertainty reported in studies?
C2	Model adequacy	Can the model reliably detect and assign probabilities to different levels of pressure?
H3a	Main hypothesis	Does adding published evidence improve early forecasts when outcome data from the current episode are few?
H3b	Safety check	Can the model reduce reliance on published evidence when it conflicts with local outcomes?
H3c	Secondary question	Do short-term changes in variability add useful predictive information?

	Role	Statement
H4	Decision value	Does any gain in forecast quality change an operational decision for the better?

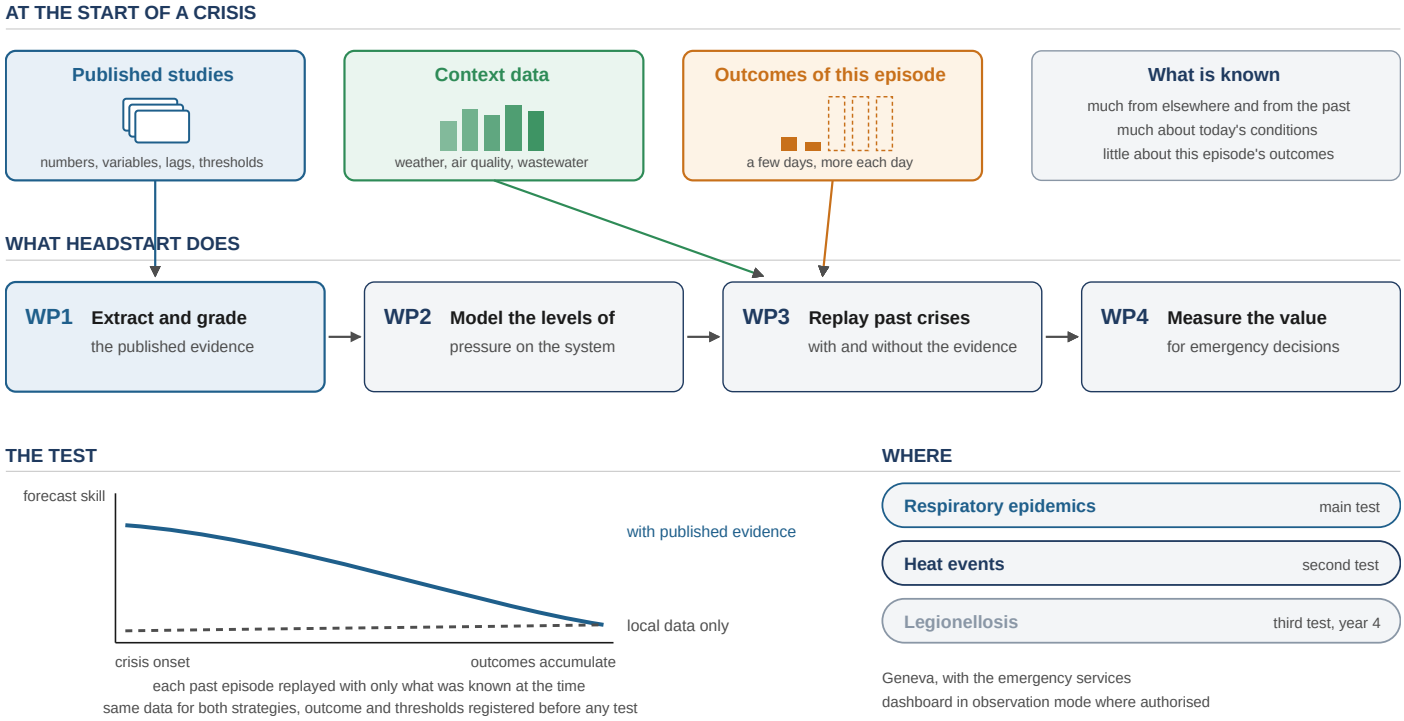


Figure 1. What exists at the start of a crisis, what HEADSTART does (WP1 to WP4), the test, and where it runs.

O1. Make published evidence usable while acknowledging its uncertainties

H1. When numbers are extracted from studies automatically, part of the uncertainty that the studies report is lost, both the uncertainty inside each study and the differences between studies. The distributions of plausible parameter values then come out too narrow. Adding a correction for extraction error should widen them back to a realistic size. To test this, a sample of studies is extracted manually and used as the reference. The automated results, with and without the correction, are compared with this reference. The correction is validated if the corrected distributions match the manual ones.

Automated extraction also misses numbers entirely [14]. H1 tests whether these omissions narrow the distributions further and whether the correction restores a useful level of uncertainty. The benchmark is built on parameter values. The other kinds of evidence (variables, model forms, thresholds) are compared with published reviews instead.

O2. Represent escalation in a form that separates the state from the point forecast

C2. Model adequacy criterion. The model must give levels of pressure that make sense to the services and probabilities that are right on average, at the same false-alarm rate for every model. This common scale lets all candidate models be compared in operational terms. It does not assume that the regime-switching model beats simpler threshold methods.

Simulated data first test whether the levels and the moves between them can be recovered from the available data. Forecast calibration is the second check. If the model fails either check, the simpler ordinal model registered as fallback is used and the main comparison still goes ahead.

O3. Test whether published evidence improves forecasts and identify instances where it is detrimental

H3a. In the first days of an episode, a strategy that uses published studies (for parameter values, explanatory variables, lags and model forms) will forecast better than a reference strategy that uses the same context data, local history and outcomes seen so far, but nothing from the literature. How long this advantage lasts as outcomes come in is a secondary question.

H3b. Adaptive weighting lowers the weight of the published evidence when local outcomes contradict it. When the evidence is right, adaptive weighting should do no worse than fixed weighting, within a margin Δ. When the evidence is deliberately made wrong, adaptive weighting should do better. This checks that the model limits harm instead of trusting the literature blindly. Δ is half of the smallest improvement counted as meaningful under H3a, so adaptive weighting may give up at most half of that benefit when the evidence is right.

H3c. Two simple signals, the recent variability of demand and how strongly each day resembles the day before, are tested as extra predictors. The question is whether they add anything to the context data, the local level and trend, and the published evidence.

Primary comparison. Both strategies choose their models with the same procedure and are scored with the continuous ranked probability score (CRPS), which rewards forecasts that put high probability near what actually happened. The main test uses respiratory episodes in their first days. The heat test is always run, but it counts as confirmatory only if the respiratory test passes, which keeps the overall false-positive rate under control. A secondary analysis follows how the difference evolves as local outcomes come in.

O4. Establish whether predictive improvement matters for decision-making

H4. The forecast with the best score is not always the most useful one for the services. The models are therefore also judged on the decisions they would trigger, using the costs of acting too early, too late or for nothing, as stated by emergency-service experts. An evidence-informed forecast counts as useful only if it changes a decision in a way that matters in practice.

Scope. Three Geneva crisis types

HEADSTART covers three Geneva crisis types. (1) Respiratory epidemics (SARS-CoV-2, influenza and RSV, together or one of them) are the main test because the published evidence is comparatively rich. (2) Heat events are the second test, with ozone and PM10 as co-exposures or effect modifiers rather than separate domains. (3) Legionellosis is the third test, in year four, on the linked case-environment data, and forecasts case incidence rather than demand. No fourth domain is planned. The outcome, the population and the pathogen of each test are not fixed in this plan. They will depend on what the published evidence defines and on the data available, for example emergency calls filtered on the recorded symptoms, emergency-department visits, hospital admissions, bed or ICU occupancy, or case counts [39, 40]. The candidates are listed at registration. The target is then chosen on access, data quality, time resolution and usefulness for the services, before any forecast performance is compared, and it does not change afterwards. Open public series support the validation of the methods but are not presented as a substitute for local service data. A month-12 data gate checks that the selected target meets the registered criteria, otherwise the next candidate takes its place. The project is not intended to outperform established forecasting hubs when long outcome histories are available. It does not assume that published evidence will improve forecasts in every setting, and it does not promise clinical decision-making by an automated system. The dashboard, where authorised, will operate in observation mode only.

2.3.2 Work packages and methods

WP1. Turn published evidence into model inputs (M1–M20)

T1.1. Define the evidence target and register the protocol (M1–M4)

Before any extraction starts, a written protocol fixes what the project looks for in the literature. There are four kinds of evidence. (1) Numbers, the core of the work, such as how demand responds to temperature, heat days, ozone and PM10, how large and how fast surges are, how long patients stay, how full units get, and what share of patients is admitted. Transmission parameters and conversions from wastewater signals to cases come second. (2) The variables and time lags that studies report as useful for each crisis type. (3) Which model families work for which outcome and horizon [41]. (4) How studies define outcomes and warning thresholds [29]. The numbers are the subject of the extraction benchmark. The other three kinds are recorded with their source and a certainty grade and compared with published reviews.

T1.2. Build the quantitative extraction benchmark (M3–M9)

For each crisis type, a systematic review run under the registered protocol identifies the relevant publications, and these form the benchmark. A second extractor, hired for this task, and I each extract the target numbers by hand from all of them when the set is small, and otherwise from a random sample drawn to cover the different kinds of parameters, reporting quality, study designs and source types, including journal articles and dated situation reports. The sample size is fixed at registration from the size of the set. Disagreements are settled by discussion and the decision is recorded. Error can then be measured separately for the sources used only in secondary analyses. As an outside check, the automated search is compared with the study lists of published systematic reviews, and the pooled numbers with the summary estimates those reviews report. The benchmark is published openly.

T1.3. Characterise automated extraction error (M7–M14)

The automated extraction is compared with the manual benchmark on four things, the values themselves, their uncertainty, the numbers it missed, and how much studies differ from each other. The main measure for H1 is simple. For each kind of parameter, the spread of the automated distribution is divided by the spread of the manual one, and the typical ratio is reported. A ratio below one means the automated extraction made the distribution too narrow. Uncertainty inside studies and differences between studies are tracked separately and together, and the analysis is repeated with each version of the extraction model.

T1.4. Construct uncertainty-aware evidence distributions (M12–M18)

Extraction errors are treated as measurement error and carried into the pooled distributions. Several ways of pooling studies are compared, weighting by precision, weighting by study quality, meta-analytic-predictive priors and power priors. Each resulting distribution keeps a record of its sources, the extraction uncertainty and the weight given to the outside evidence.

T1.5. Transportability screen (M15–M20)

Each finding is screened for how well it transfers to the local setting, looking at the factors that change an effect and at the differences between the study setting and the local one. The screening criteria, and how a low score widens the distribution or lowers the weight of the evidence, are fixed before the forecast comparison starts.

Deliverables are D1.1 (open extraction benchmark), D1.2 (extraction-error analysis) and D1.3 (evidence-derived model-input library).

Risk and mitigation. If manual extraction time is limited, the benchmark narrows to the core kinds of parameters.

WP2. Model levels of health-system pressure (M1–M28)

T2.1. Specify and identify the state model (M1–M9)

The model follows how pressure on the system moves between ordered levels, for example routine, elevated, strained and critical, and gives the probability of each level for each day. It is a Bayesian regime-switching model [42]. Its main input is the registered outcome, with other outcomes such as emergency-department visits and ICU occupancy added when access allows. Weather, air quality, calendar effects and epidemic indicators are the explanatory variables. The model is first tested on simulated data to check that it can find the levels and the moves between them, also with a single data series. If it cannot, a simpler ordinal model is used. The number of levels and their thresholds are fixed at registration, from the literature where it gives usable definitions. Otherwise four default levels are set on seasonal percentiles (routine below the 75th, elevated to the 90th, strained to the 97.5th or when capacity is under sustained pressure, with ICU occupancy above 85% or emergency-department boarding above its seasonal 90th percentile, and critical above the 97.5th or when capacity is saturated). The highest level that applies counts. These thresholds guide the model and are not hard cut-offs. Experts can later refine what each level means for decisions, not the definitions used in the main test.

T2.2. Represent the critical tail (M6–M14)

Rare, very high demand is modelled with extreme-value methods (peaks over a threshold, generalised Pareto distribution) and linked to the probability of the critical level. The sensitivity to the chosen threshold is reported. This work supports the main comparison and does not carry a hypothesis of its own.

T2.3. Include published evidence in the model parameters (M10–M20)

The distributions built in WP1 feed the model parameters for which published evidence exists, such as weather effects, surge size and the moves between levels. Three ways of using them are compared. The first uses almost no outside information. The second uses the evidence distributions as they are. The third lowers the weight of the evidence when a registered conflict measure shows that local outcomes disagree with it. The conflict measure and the weight rule are fixed before evaluation. Power priors are run as a sensitivity check.

T2.4. Add resilience indicators as a secondary information channel (M12–M20)

The two signals of H3c, the recent variability of demand and how much each day resembles the day before, are tested as optional predictors of a move to a higher level. The question is whether they add anything to the context data, the recent level and trend, and the published evidence. A negative answer is also a useful result.

T2.5. Calibration, automation and validity monitoring (M20–M28)

Forecast probabilities are calibrated with methods made for data that change over time. Public feeds (weather, air quality, wastewater, surveillance indicators) are loaded automatically as often as they are updated. Partner data are linked only under access agreements. The pipeline records when each value was observed and released, which version of the data and of the model was used, refits the models on a fixed schedule and checks after each fit that the residuals, the spread and the calibration are acceptable. Models that fail these checks are set aside. The working code is released within LiteRev-Evidence.

Simulated data are used only to test the model, never as extra observations. Treating simulations based on the literature as real outcomes would count the same information twice and spoil the test of conflict between the evidence and the local data.

Deliverables are D2.1 (model specification and identifiability study), D2.2 (open implementation) and D2.3 (methods paper).

Risk and mitigation. If the states cannot be separated reliably, the registered ordinal model is used and the identifiability limit is reported.

WP3. Test forecasts using historically available information (M12–M42)

T3.0. Outcomes, data gate and episode eligibility (M1–M14)

Outcomes. The outcome of each test is not fixed in advance. It depends on what the published evidence defines and on the data available, for example emergency calls (CASU-144) filtered on the symptoms recorded for each call, emergency-department visits by cause, hospital admissions, bed or ICU occupancy, the HUG COVID-19 hospitalisation series I analysed before [35], or well-known open series. The candidates are listed at registration in order of preference. The target is chosen on access, data quality, time resolution and usefulness for the services, before any forecast is compared. The causes and population groups used to filter it come from the evidence. Once chosen, the target stays fixed, so that no outcome can be picked afterwards because it gives better results. The other candidates serve as extra inputs for the pressure model. Weather, air quality, wastewater, sentinel consultations and calendar effects are context inputs. Open public series help validate the methods but do not replace local service data. The design is single-canton on purpose, and transfer is tested across the three crisis types and on the open national benchmark.

Data gate. At month 12 the chosen target is checked against criteria on history length, resolution, reporting delay and completeness fixed in advance. If it fails, the next outcome in the list takes its place. Performance plays no part in this choice.

Episode eligibility. An episode enters the main comparison only if its start can be identified without later information, it has enough history before it and enough outcomes after it, the literature available at each forecast date can be rebuilt, and it can be separated from neighbouring episodes. Other episodes serve for sensitivity analyses. Respiratory viruses circulating at the same time count as one episode.

Registration. The outcome list, the gate and the eligibility rules are registered first. After the month-12 check, but before any performance is seen, a second registration fixes the early-episode window N, the main forecast horizon per crisis type and the margin Δ. Horizons follow what the services need, not results.

Open-data track (M12–M24). Before clinical records are connected, the whole pipeline runs on open Swiss series (federal respiratory surveillance, national wastewater data, weekly deaths and MeteoSwiss weather), with archived multi-model forecasts as comparators [3, 4]. This checks that the replay and selection machinery works.

T3.1. Assemble the retrospective information set (M12–M20)

The outcome series are lined up with the context feeds and with each other on a daily grid. Missing values and reporting delays are measured, and recent days that are still incomplete are corrected with nowcasting [43, 44], because reporting tends to slow down under pressure and can look like an early warning. The HUG COVID-19 hospitalisation data I analysed before [35] are requested again as an extra series to check and calibrate the respiratory models. Pharmacy sales and wastewater

measurements are requested through the cantonal pharmacist and used as early indicators where access and coverage allow.

T3.2. Reconstruct the true information set (M18–M30)

For each past episode, forecasts are made at successive dates using only what was known on that date. The evidence comes from papers and preprints published before each date. Dated WHO and ECDC reports about events elsewhere are used only in a secondary analysis that tests a wider evidence feed. Any text describing the local episode itself is excluded, because it could give away the outcome. Every input carries two dates, when it was observed and when it became available. When a source was later corrected, the replay uses the version available at the time, not the corrected one. Automatic feeds are replayed at their own frequency. The candidate horizons are 7, 14 and 28 days, and the main horizon is fixed at the second registration on what the services need. The early window of each episode starts at a rule that uses only information known at the time.

T3.3. The model library and the automated selection engine (M20–M34)

Two strategies are registered before evaluation. The reference strategy uses a registered set of standard forecasting models (seasonal regression, persistence, surveillance alarm methods [45], machine-learning learners and the WP2 pressure model with almost no outside information), fitted to the local history, the outcomes already seen in the episode and the context feeds, with nothing taken from the literature. The evidence-informed strategy uses the same inputs plus the WP1 distributions and the registered advice on variables, lags and model forms, in three versions, fixed evidence weights, adaptive weights, and adaptive weights plus the two signals of H3c. Transmission models are allowed only for infectious crises. The allowed candidates are fixed per crisis type at registration.

At each forecast date, each strategy picks its best candidate using only outcomes seen before that date, with the CRPS as the ranking criterion and log score and calibration as entry checks. The chosen model then forecasts the following period, whose outcomes played no part in the choice. Both strategies follow the same rules. The main comparison tests the whole evidence-informed strategy, and secondary comparisons separate the effect of the priors, of the variable and lag advice, and of the model forms.

The tests run in a fixed order. Test 1 compares the two strategies on the CRPS skill score over the early window of respiratory episodes, two-sided at $\alpha = 0.05$, because harm matters as much as benefit. Success needs a paired permutation p-value below α and a lower confidence bound above the registered minimum improvement. Test 2 repeats the comparison for heat. It is confirmatory if Test 1 passes and exploratory otherwise, and if Test 1 fails H3a is not supported. The reference models and their settings are fixed at registration so that the comparator cannot be weakened later.

Main endpoint. For each episode and strategy, the CRPS is averaged over the forecast dates in the window at the main horizon, and the skill score is one minus the ratio of the two averages, so that positive values favour the evidence-informed strategy. The main statistic is the mean over episodes, the episode being the unit of analysis because only 13 to 14 candidate respiratory episodes exist. A paired permutation test and a bootstrap over whole episodes are both registered. Simulations run before the second registration set the type-I error, the power, the smallest number of episodes for a confirmatory claim and the window N, without using any forecast result. The minimum relevant improvement is fixed at the second registration by a rule based on operational relevance, and Δ is half of it. Secondary measures are the log score, calibration and the detection of escalation at matched false-alarm rates. For H3b, adaptive weighting passes when its skill score is no more than Δ below fixed weighting with correct evidence and better than fixed weighting with shifted evidence.

T3.4. Map benefit and failure (M28–M38)

The analysis finds the episodes where the published evidence helped and those where it hurt, and looks for the reasons, such as differences in population, outcome definition, service organisation, policy or period, extraction uncertainty, or conflict with the local outcomes. Deliberately shifted evidence serves as a stress test. A key question is whether adaptive weighting reduces harm early enough. The result is a map of when the approach helps, when it does not, and why, not a single average.

T3.5. The second and third tests (M34–M42)

The heat test is the second test. The legionellosis test, in year four, applies the same comparison to case-incidence forecasts at the start of an outbreak, using the linked case-environment data.

Deliverables are D3.1 (reproducible historical forecast-evaluation pipeline), D3.2 (primary result), D3.3 (benefit and failure analysis) and D3.4 (analysis across crisis types).

Risk and mitigation. Access to operational records may be delayed. Agreements are being pursued before the award. The open-data track can validate the reconstruction and evaluation methods without clinical data, but it cannot replace the main test on local service data. If access fails, the scientific claim narrows and the decision analysis is limited accordingly.

WP4. From predictive skill to operational value (M24–M48)

T4.1. Elicit operational losses and thresholds (M24–M32)

A structured elicitation (SHELF method) with two to five experienced people from HUG emergency medicine, CASU-144 regulation and capacity management puts numbers on the cost of escalating too early, too late or for nothing. The decision thresholds come from these costs. The decision studied is fixed in advance, namely escalating surge capacity when the forecast probability of the upper levels (strained or critical in the default scheme) passes the threshold. Models that forecast counts rather than levels are converted to the same level probabilities by a registered rule.

T4.2. Decision-analytic evaluation and fairness audit (M30–M40)

The WP3 forecasts are re-assessed on the decisions they would trigger, with net-benefit and decision-curve methods, to see whether the ranking of the models changes once the costs of wrong decisions are counted. Calibration and threshold performance are also checked across groups by age and sex. A model that is well calibrated overall but poor for a group that matters for decisions does not pass.

T4.3. Observation-mode dashboard and living updating (M34–M48)

Where a strategy passes the forecast and decision checks and the services authorise it, the project runs a daily dashboard in observation mode for that crisis type. It publishes forecasts, level probabilities and validity checks, records its outputs and scores them as outcomes arrive, and it does not direct clinical care. New studies are checked daily, and full syntheses are refreshed on a registered schedule [13]. Any change to priors, variables, thresholds or model choice is version-controlled and applied only under pre-registered rules. Without authorisation, the retrospective evaluation and the update protocol remain the deliverables.

Deliverables are D4.1 (cost structure and fairness audit), D4.2 (decision-analytic evaluation) and D4.3 (the update protocol and, where authorised, the dashboard).

Reproducibility and data protection. Code is version-controlled in R and Python under a registered analysis plan. Clinical data stay in the UNIGE secure environment under the applicable approvals. The benchmark and the software are released openly, with synthetic data where real data cannot be shared.

The project delivers peer-reviewed papers and reusable resources (the extraction benchmark, the public forecasting benchmark, the library of evidence-derived inputs and the decision-analysis framework), and reports its results whether the evidence helps, has no effect or causes harm.

2.3.3 Research environment, team and resources

The project is based at the Institute of Global Health (ISG), Faculty of Medicine, University of Geneva, which brings together epidemiology, infectious-disease modelling and evidence synthesis. The programme is led independently alongside the Institute's groups.

Entity	Person in charge	Role in HEADSTART	Status
ISG, UNIGE	Prof. Olivia Keiser, contact person	Host institute	Hosting agreed, SNSF letters in preparation
DS4DH, UNIGE	Prof. Douglas Teodoro	Biomedical text mining for WP1	Letter of support signed
HUG, emergency department	Prof. Thibaut Desmettre, head of department	ED and ICU data, clinical partner	Extraction in discussion, letter requested
CASU-144, HUG	Dr Robert Larribau	Emergency-call data, clinical partner	Access held within GESICA, project agreement in preparation
GESICA (Franco-Swiss)	Prof. Olivia Keiser (UNIGE PI), Prof. Douglas Teodoro (Swiss data inventory)	Surveillance inventory, data access, emergency-medicine partners	Running
GeoAI4EI (Horizon Europe)	Prof. Olivia Keiser, PI	European epidemic-intelligence partners, exchange of methods	Running 2026–2030
Médecin cantonal and SIG		Legionellosis linked data	Secured, BASEC 2026-00324
Pharmacien cantonal		Wastewater and pharmacy sales	Contacts established
UNIGE HPC and secure data environment		Computing	In place
Open public series		Surveillance, wastewater, deaths, weather	Public

The ED and ICU extraction and the project-specific CASU-144 agreement are not yet secured. Each rests on an existing working relationship.

Project funds are capped at CHF 250,000 over four years, and my salary is separate.

Item	Rationale
Research assistants (master's students, at most 40%)	Second extraction arm of the T1.2 benchmark (M3–M9), data handling, evaluation pipeline, workshop organisation
Database and dashboard services	Set-up of the project database, secure-environment runs and, where authorised, dashboard hosting

Item	Rationale
Conferences, two to three per year, and workshops	Presentation of results, exchange with the forecasting hubs and the GESICA and GeoAI4EI partners
Other eligible direct costs	As justified in the final SNSF budget

Before month 1 and outside the grant, I will finalise the data agreements with HUG and CASU-144, confirm data reuse under the applicable governance rules, and harden LiteRev-Evidence before clinical data are connected.

2.4 Schedule and milestones

I lead all four work packages. Research assistants (master's students) provide the second extraction arm from months 3 to 9 and support data handling and the evaluation pipeline. The work packages run in parallel with fallbacks. WP2 can use weakly informative distributions if extraction error proves too large, the open-data track runs without clinical records, and the retrospective decision analysis of WP4 does not depend on live deployment. The open-data track cannot replace the test on local service data, so access agreements remain a priority before the award.

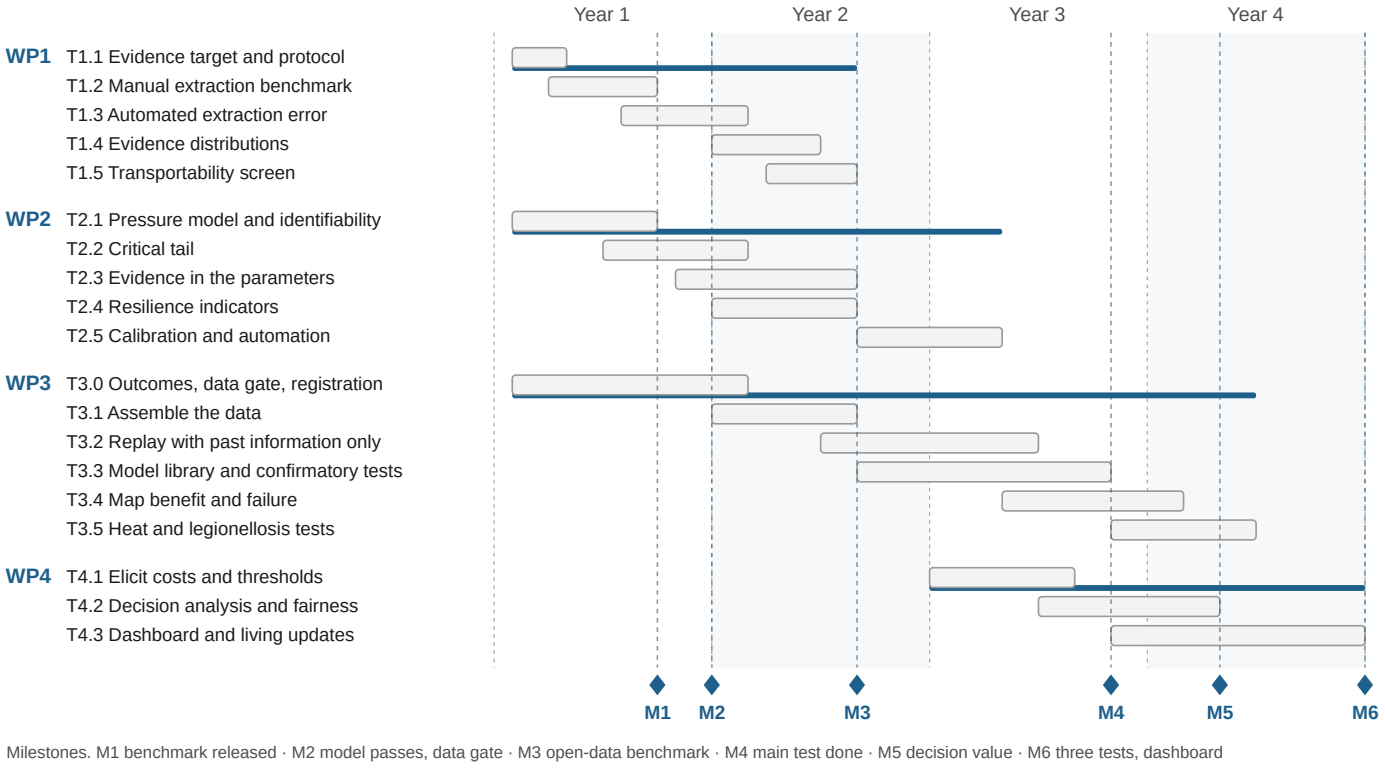


Figure 2. Work packages, task durations and milestones across the 48-month project.

Milestone	Month	Criterion
M1	9	Extraction benchmark released
M2	12	Pressure model passes the identifiability checks, or fallback chosen
M3	20	Open-data benchmark running, service series harmonised (or fallback)
M4	34	Main hypothesis tested by the forecast comparison
M5	40	Decision relevance established
M6	48	Three crisis types analysed, dashboard running where authorised

2.5 Relevance and impact

If published evidence does not improve forecasts, that is still a useful result. It shows where using published estimates as model inputs adds little or carries too much risk, and spares modellers and evidence platforms an untested assumption.

Results are published in open-access journals, with preprints where appropriate. The extraction benchmark, the forecasting benchmark, the library of evidence-derived inputs and the code are released as versioned, citable resources, and the findings go back to the participating services. The results and tools are also shared with GESICA and GeoAI4EI, which can carry them to the Franco-Swiss and European epidemic-intelligence networks.

The project will show whether outside evidence improves forecasts in the first weeks of a crisis, under which conditions, and with how much certainty. For the crisis types that pass validation and are authorised, it delivers a registered forecasting configuration and a dashboard in observation mode. Wider deployment to other crises is a possible next step after the grant, not a promised outcome.

2.6 Relevance for personal career development

I built the methods, the prototype and the data relationships that make this project feasible, mostly inside existing programmes. HEADSTART gives me my own scientific question and a programme that I lead.

Programme	Objective	My role	Distinction from Ambizione
GESICA	Franco-Swiss crisis intelligence	Contributor, data sourcing for respiratory diseases	Decision support, no test of evidence-derived priors
GeoAI4EI (Horizon Europe)	European epidemic-intelligence toolbox	Contributor	Infrastructure, no test of borrowed evidence
Legionellosis study (ongoing)	Geneva waterborne outbreak determinants	Full analysis	Supplies the data of the third test
Ambizione	When published evidence improves forecasts, and when it misleads	PI, full-time	-

GESICA and GeoAI4EI build epidemic-intelligence and decision-support tools. HEADSTART tests one thing they do not, whether published evidence is a sound input to forecasts.

LiteRev-Evidence can already turn published estimates into parameter distributions, but nobody has shown whether using them improves forecasts in a new setting. That question, the design that answers it and the confirmatory analyses are mine.

Prof. Olivia Keiser, my contact person at the Institute, co-built LiteRev with me, leads the legionellosis study and is my PI in GESICA (respiratory diseases) and in GeoAI4EI (Horizon Europe). We also work together on the MAS in Public Health, where I have taught statistics and epidemiology since 2024. Prof. Douglas Teodoro (DS4DH) leads the data inventory of the Swiss part of GESICA and supports the text-mining work of WP1. HEADSTART is my own programme, hosted at the Institute of Global Health and led by me, with research assistants on defined technical tasks. By month 48, a validated answer and a first-author record will be the basis for leading a group or a professorship.

Bibliography

1. Farrington CP, Andrews NJ, Beale AD, Catchpole MA. A statistical algorithm for the early detection of outbreaks of infectious disease. *Journal of the Royal Statistical Society Series A* 1996;159(3):547–563.
2. Noufaily A, Enki DG, Farrington P, Garthwaite P, Andrews N, Charlett A. An improved algorithm for outbreak detection in multiple surveillance systems. *Statistics in Medicine* 2013;32(7):1206–1222. doi:10.1002/sim.5595
3. Cramer EY, Ray EL, Lopez VK, et al. (consortium, more than 50 authors). Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States. *Proceedings of the National Academy of Sciences* 2022;119(15):e2113561119. doi:10.1073/pnas.2113561119
4. Sherratt K, Gruson H, Grah R, et al. (consortium, more than 50 authors). Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. *eLife* 2023;12:e81916. doi:10.7554/eLife.81916
5. Rosenkötter N, Ziemann A, Garcia-Castrillo Riesgo L, Gillet JB, Vergeiner G, Krafft T, Brand H. Validity and timeliness of syndromic influenza surveillance during the autumn/winter wave of A(H1N1) influenza 2009: results of emergency medical dispatch, ambulance and emergency department data from three European regions. *BMC Public Health* 2013;13:905. doi:10.1186/1471-2458-13-905
6. Bonora R, Ticozzi EM, Pregliasco FE, Pagliosa A, Bodina A, Cereda D, Perotti G, Lombardo M, Stirparo G. Telephone calls to emergency medical service as a tool to predict influenza-like illness: a 10-year study. *Public Health* 2025;238:239–244. doi:10.1016/j.puhe.2024.12.021
7. Edjinedja KL, Larribau R, Orel E, Cossus J, Elfahim O, Yendouname K, Vaussenat F, Teodoro D, Robert-Nicoud S, Barakat O, Desmettre T. Artificial intelligence in emergency medical services for disasters and health emergencies: a systematic review. Submitted, 2026.
8. Lazer D, Kennedy R, King G, Vespignani A. The parable of Google Flu: traps in big data analysis. *Science* 2014;343(6176):1203–1205. doi:10.1126/science.1248506
9. Ibrahim JG, Chen M-H. Power prior distributions for regression models. *Statistical Science* 2000;15(1):46–60.
10. Hobbs BP, Carlin BP, Mandrekar SJ, Sargent DJ. Hierarchical commensurate and power prior models for adaptive incorporation of historical information in clinical trials. *Biometrics* 2011;67(3):1047–1056.
11. Schmidli H, Gsteiger S, Roychoudhury S, O'Hagan A, Spiegelhalter D, Neuenschwander B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics* 2014;70(4):1023–1032. doi:10.1111/biom.12242
12. Cook JD, Williams DM, Walsh DP, Hefley TJ. Bayesian forecasting of disease spread with little or no local data. *Scientific Reports* 2023;13:8137. doi:10.1038/s41598-023-35177-6
13. Elliott JH, Turner T, Clavisi O, Thomas J, Higgins JPT, Mavergames C, Gruen RL. Living systematic reviews: an emerging opportunity to narrow the evidence-practice gap. *PLoS Medicine* 2014;11(2):e1001603. doi:10.1371/journal.pmed.1001603
14. Shankar R, Lim A, Qian X. Performance of large language models in data extraction for evidence synthesis: a systematic review. *Journal of Biomedical Informatics* 2026;181:105086. doi:10.1016/j.jbi.2026.105086

15. Bareinboim E, Pearl J. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences* 2016;113(27):7345–7352. doi:10.1073/pnas.1510507113
16. Dahabreh IJ, Petito LC, Robertson SE, Hernán MA, Steingrimsson JA. Toward causally interpretable meta-analysis: transporting inferences from multiple randomized trials to a new target population. *Epidemiology* 2020;31(3):334–344. doi:10.1097/EDE.0000000000001177
17. Degtiar I, Rose S. A review of generalizability and transportability. *Annual Review of Statistics and Its Application* 2023;10:501–524.
18. Le Strat Y, Carrat F. Monitoring epidemiologic surveillance data using hidden Markov models. *Statistics in Medicine* 1999;18(24):3463–3478.
19. Watkins RE, Eagleson S, Veenendaal B, Wright G, Plant AJ. Disease surveillance using a hidden Markov model. *BMC Medical Informatics and Decision Making* 2009;9:39. doi:10.1186/1472-6947-9-39
20. Coles S. *An Introduction to Statistical Modeling of Extreme Values.* Springer, 2001. doi:10.1007/978-1-4471-3675-0
21. Ranjbar S, Cantoni E, Chavez-Demoulin V, Marra G, Radice R, Jaton-Ogay K. Modelling the extremes of seasonal viruses and hospital congestion: the example of flu in a Swiss hospital. *Journal of the Royal Statistical Society Series C (Applied Statistics)* 2022;71(4):884–905. doi:10.1111/rssc.12559
22. Keeling MJ, Rohani P. *Modeling Infectious Diseases in Humans and Animals.* Princeton University Press, 2008.
23. Scheffer M, Bascompte J, Brock WA, Brovkin V, Carpenter SR, Dakos V, Held H, van Nes EH, Rietkerk M, Sugihara G. Early-warning signals for critical transitions. *Nature* 2009;461:53–59. doi:10.1038/nature08227
24. O'Regan SM, Drake JM. Theory of early warning signals of disease emergence and leading indicators of elimination. *Theoretical Ecology* 2013;6:333–357.
25. Brett TS, O'Dea EB, Marty É, Miller PB, Park AW, Drake JM, Rohani P. Anticipating epidemic transitions with imperfect data. *PLoS Computational Biology* 2018;14(6):e1006204. doi:10.1371/journal.pcbi.1006204
26. Southall E, Brett TS, Tildesley MJ, Dyson L. Early warning signals of infectious disease transitions: a review. *Journal of the Royal Society Interface* 2021;18(182):20210555. doi:10.1098/rsif.2021.0555
27. Gneiting T, Raftery AE. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 2007;102(477):359–378.
28. Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making* 2006;26(6):565–574.
29. Lung S-CC, Yeh J-CJ, Hwang J-S. Selecting thresholds of heat-warning systems with substantial enhancement of essential population health outcomes for facilitating implementation. *International Journal of Environmental Research and Public Health* 2021;18(18):9506. doi:10.3390/ijerph18189506
30. Zheng XY, Ding H, Jiang LN, Chen SW, Zheng JP, Qiu M, Zhou YX, Chen Q, Guan WJ. Association between air pollutants and asthma emergency room visits and hospital admissions in time series studies: a systematic review and meta-analysis. *PLoS ONE* 2015;10(9):e0138146. doi:10.1371/journal.pone.0138146

31. Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa. *PLoS ONE* 2022;17(3):e0264429. doi:10.1371/journal.pone.0264429
32. Merzouki A, Estill J, Orel E, Tal K, Keiser O. Clusters of sub-Saharan African countries based on sociobehavioural characteristics and associated HIV incidence. *PeerJ* 2021;9:e10660. doi:10.7717/peerj.10660
33. Esra R, Carstens J, Le Roux S, Mabuto T, Eisenstein M, Keiser O, Orel E, Merzouki A, De Voux L, Maskew M, Sharpey-Schafer K. Validation and improvement of a machine learning model to predict interruptions in antiretroviral treatment in South Africa. *Journal of Acquired Immune Deficiency Syndromes* 2023;92(1):42–49.
34. Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. An automated literature review tool (LiteRev) for streamlining and accelerating research using natural language processing and machine learning. *Journal of Medical Internet Research* 2023;25:e39736. doi:10.2196/39736
35. Orel E, Sobel J, Zanella MC, Iten A, Kaiser L, Keiser O, Vetter P. Clinical severity of Delta and 3 different Omicron sublineages in patients hospitalized because of COVID-19 in a Swiss tertiary center. *CMI Communications* 2024;1(2):100005. doi:10.1016/j.cmicom.2024.100005
36. Estill J, Venkova-Marchevska P, Roelens M, Orel E, Temerev A, Flahault A, Keiser O. Future scenarios for the SARS-CoV-2 epidemic in Switzerland: an age-structured model. *F1000Research* 2020;9.
37. Nwosu KD, Fokam J, Wanda F, Mama L, Orel E, Ray N, Meke J, Tassegning A, Takou D, Mimbe E, Stoll B, Guillebert J, Comte E, Keiser O, Ciaffi L. SARS-CoV-2 antibody seroprevalence and associated risk factors in an urban district in Cameroon. *Nature Communications* 2021;12:5851. doi:10.1038/s41467-021-25946-0
38. Ng'ambi W, Estill J, Keiser O, Merzouki FA, Orel E. Machine learning-based classification of self-reported cardiovascular disease history in Africa using harmonised multi-country WHO STEPS surveys: 2014–2019. Accepted for publication.
39. Schulte F, Röösli M, Ragettli MS. Risk, attributable fraction and attributable number of cause-specific heat-related emergency hospital admissions in Switzerland. *International Journal of Public Health* 2024;69:1607349. doi:10.3389/ijph.2024.1607349
40. Ragettli MS, Vicedo-Cabrera AM, Schindler C, Röösli M. Impact of the warm summer 2015 on emergency hospital admissions in Switzerland. *Environmental Health* 2019;18:66. doi:10.1186/s12940-019-0507-1
41. Wargon M, Guidet B, Hoang TD, Hejblum G. A systematic review of models for forecasting the number of emergency department visits. *Emergency Medicine Journal* 2009;26(6):395–399. doi:10.1136/emj.2008.062380
42. Hamilton JD. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 1989;57(2):357–384.
43. Höhle M, an der Heiden M. Bayesian nowcasting during the STEC O104:H4 outbreak in Germany, 2011. *Biometrics* 2014;70(4):993–1002. doi:10.1111/biom.12194
44. McGough SF, Johansson MA, Lipsitch M, Menzies NA. Nowcasting by Bayesian smoothing: a flexible, generalizable model for real-time epidemic tracking. *PLoS Computational Biology* 2020;16(4):e1007735. doi:10.1371/journal.pcbi.1007735

45. Salmon M, Schumacher D, Höhle M. Monitoring count time series in R: aberration detection in public health surveillance. *Journal of Statistical Software* 2016;70(10):1–35. doi:10.18637/jss.v070.i10